Niketh Prakash Achanta

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2028	8.10
Shri Ram Global School, Whitefield Bangalore	Class XII (CBSE)	2024	95.20%
Delhi Public School, Bangalore East	Class X (CBSE)	2022	96.80%

SKILLS

Languages: C, Python, Verilog, MATLAB

EDA Tools: Cadence Virtuoso, Intel Quartus Prime, LTspice, Altium Designer

Other Tools/Skills: Linux, Git, LaTex, Arduino IDE

PROJECTS

8-bit Breadboard Computer

- Built an 8-bit computer using discrete TTL logic implementing a shared system bus, EEPROM-based program memory, RAM, ALU, and program counter.
- Implemented the instruction fetch-decode-execute cycle and custom machine instructions through hardwired control logic.
- Gained hands-on understanding of computer architecture, memory organization, bus arbitration, and timing synchronization.
- Debugged hardware-level timing and signal integrity issues during system integration.

Closed-Loop Stepper Valve Controller for LPCVD System

- Developing an embedded control system for regulating chamber pressure in an LPCVD furnace using closed-loop feedback.
- Interfacing a pressure sensor with a microcontroller to control valve position through a stepper motor driver.
- Implementing firmware for real-time pressure regulation and actuator control.
- Integrating sensing, motor control, and embedded software for laboratory deployment.

CMOS Level Shifter Design (Cadence Virtuoso)

- Designed a CMOS level shifter in Cadence Virtuoso for reliable voltage-domain translation between digital logic levels.
- Performed transistor-level schematic design and optimized device sizing for robust switching behavior.
- Verified functionality using Cadence Spectre simulations.

Operational Amplifier Characterization PCB

- Designed a PCB-based measurement platform for evaluating operational amplifier parameters including slew rate and gain-bandwidth product.
- Developed analog signal-conditioning circuitry and performed schematic capture, PCB layout, and component selection.
- Currently performing hardware testing and measurement verification.

Custom 4-bit Successive Approximation (SAR) ADC

- Designed a 4-bit SAR ADC using a Sample-and-Hold circuit, Comparator, R-2R Ladder DAC, and SAR control logic.
- Implemented binary-search-based analog-to-digital conversion using successive approximation.
- Characterized converter performance using Offset Error, Gain Error, DNL, and INL measurements.
- Verified sampling behavior through FFT-based frequency-domain analysis.

RELEVANT COURSES

Core Courses: Digital Systems, Analog Electronics, Signal Processing

Mathematics: Mathematics for Electrical Engineers, Matrix Theory, Differential Equations & Transform Techniques, Probability & Random Processes

Computer Science / Programming: Introduction to Programming, Data Structures & Applications, Scientific Programming for Electrical Engineers

Lab Courses: Electronic Devices & Circuits Lab, Digital Systems Lab, DSP Lab, PCB Design

POSITIONS OF RESPONSIBILITY

Elected **President, Student Gymkhana**, representing and leading the student body of a **5,000+ student** campus.

Overseeing student welfare, campus activities, and coordination with institute administration to drive student initiatives, resolve concerns, and enhance campus engagement.

Managed a **10 lakh** annual budget as **Vice-Secretary, SPIC MACAY**, ensuring effective planning and execution of cultural programs.

Coordinated performances by **Padma award-winning artists** and led event logistics to promote Indian classical arts, culture, and heritage across the campus.

Elected **Class Representative** for **two consecutive years**, representing the academic interests of the class and serving as the primary liaison between students and faculty.

EXTRACURRICULAR

Asia Rank 14 - MaRRS International Spelling Bee

Played the lead role of Genie in a **90+ minute** live production of *Aladdin* before an audience of **2,000+**